

Autoencoders for Stellar Activity Removal in AU-Mic Spectra

Marvin Michel Ernst



Practicum Presentation

Supervisor: Dr. Manuel Perger (ICE-CSIC)

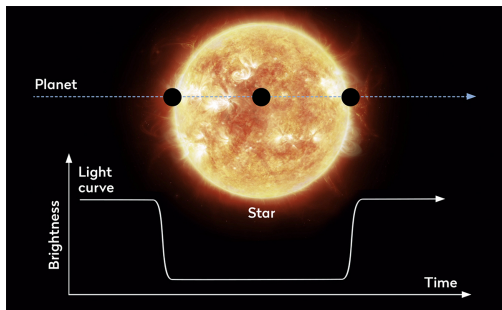
Barcelona, June 25, 2025

Overview

- 1 Motivation
- 2 Data and Preprocessing
- 3 Autoencoder Modeling
- 4 Example Configuration
- 5 Conclusion

Motivation

Exoplanet Detection via Transit Method: Photometric Observations.



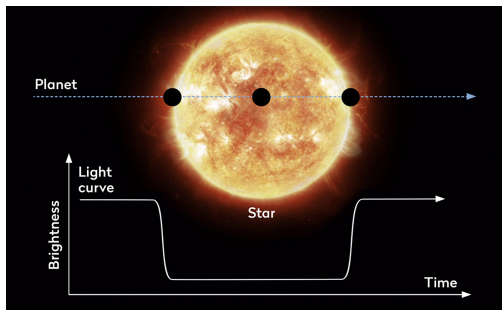
Instrumental precision: 100 ppm

Earth-twin: 10 ppm

Stellar activity: 200 - 300 ppm

Motivation

Exoplanet Detection via Transit Method: Photometric Observations.



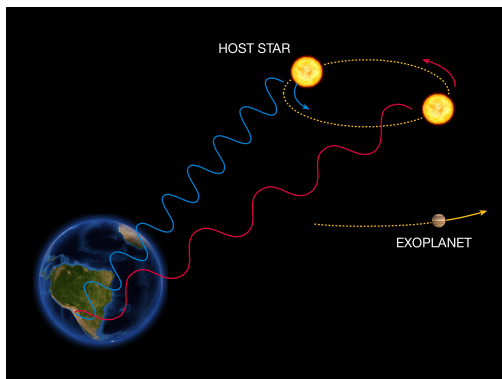
Instrumental precision: 100 ppm

Earth-twin: 10 ppm

Stellar activity: 200 - 300 ppm

Motivation

Exoplanet Detection v. Radial Velocity Method: Spectral Observations.



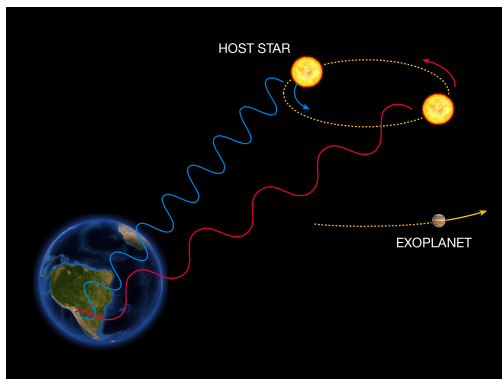
Instrumental precision: 1 m/s

Earth-twin: 10 cm/s

Stellar activity: 2-3 m/s

Motivation

Exoplanet Detection v. Radial Velocity Method: Spectral Observations.

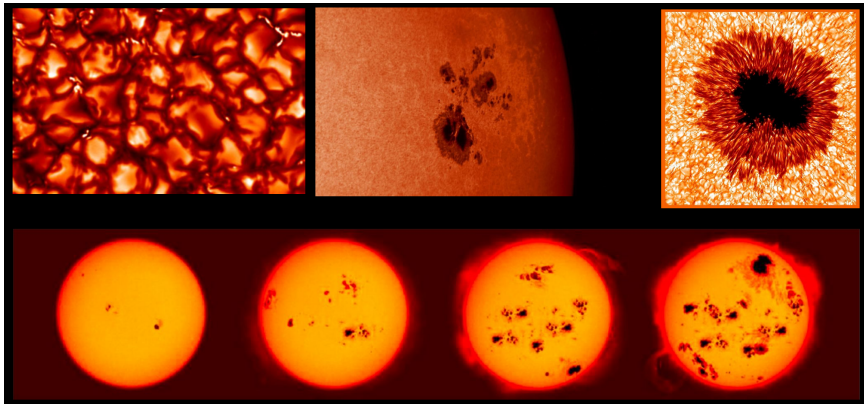


Instrumental precision: 1 m/s

Earth-twin: 10 cm/s

Stellar activity: 2-3 m/s

Problem: Stellar Noise from Magnetic Activity

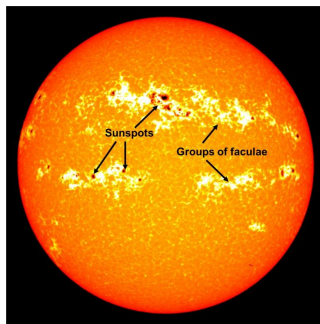


Project Goal

Objective

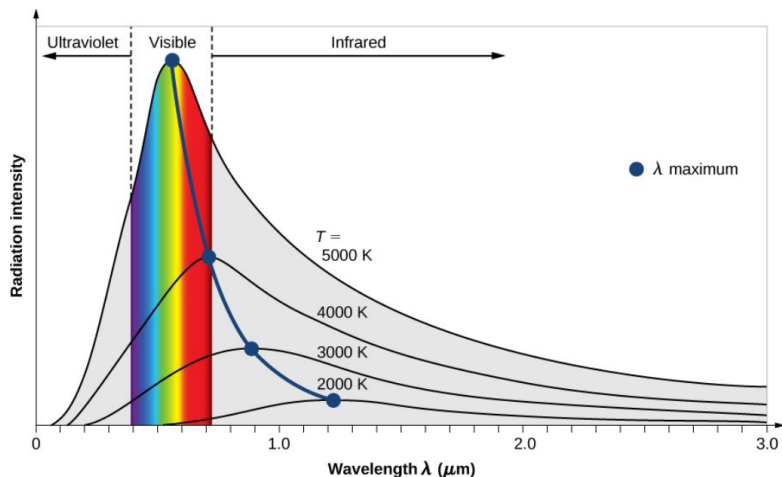
We aim to develop an additional - ideally improved - parameterization of **stellar magnetic activity**, primarily caused by *dark starspots* and *bright faculae*.

This activity contaminates all efforts to **detect** and **characterize** exoplanets and their atmospheres.

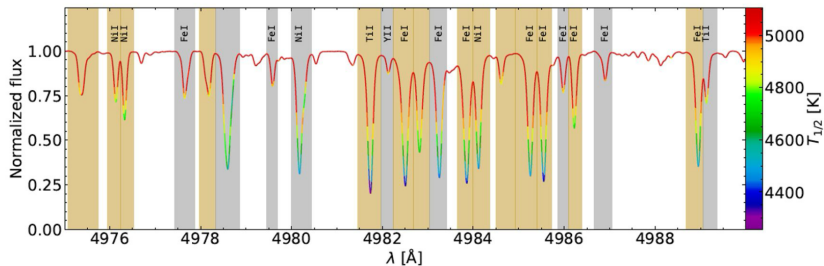


Data and Preprocessing

Data: Stellar Light - Radiation Curves

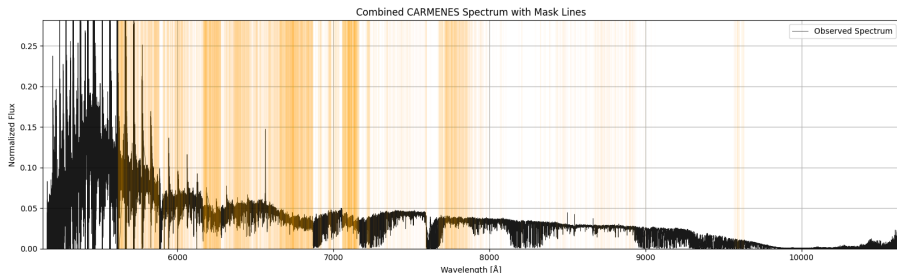


Data: Stellar Light - High-Resolution Spectrum



Data: AU Mic Spectra from CARMENES

- **Target:** AU Microscopii (AU Mic), an active M dwarf with strong magnetic variability and known planets.
- **Dataset:** 101 spectra among two observation periods
Each spectrum contains:
 - 61 spectral orders
 - 4096 flux values per order
- **Mask:** Absorption lines are dips in stellar spectra.

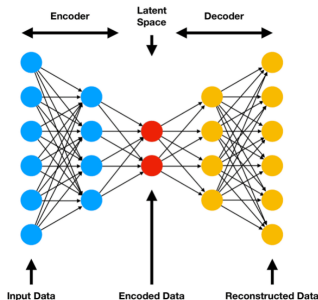


Modeling Approach: Neural Network

Core Idea

Use an **autoencoder** to learn a compressed representation that captures the dominant sources of spectral variability.

- **Input:** Preprocessed spectra from AU Mic.
- **Encoder:** Compresses each spectrum into a *latent vector*.
- **Latent space:** Low-dimensional representation.
- **Decoder:** Reconstructs spectrum from latent code.



Data Preprocessing: Overview

- **Use telluric-corrected spectra.**
- Remove noisy spectra with SNR deviating by more than 3σ from the mean $\rightarrow$ 95 spectra (before 101).
- Continuum normalization to flatten large-scale flux variations.
- **Optional preprocessing steps:**
 - Radial velocity correction using the relativistic Doppler formula:

$$\lambda_{\text{shifted}} = \lambda \cdot \sqrt{\frac{1 + v/c}{1 - v/c}} \quad \text{where} \quad v = v_{\text{RV}}$$

- Zero-mean normalization per spectrum: subtract mean flux.
 - Unit-variance scaling per spectrum: divide by global SD.
 - Region selection: specific spectral orders or wavelength intervals.
- All spectra are interpolated to a common wavelength grid.

Data Preprocessing: Overview

- Use telluric-corrected spectra.
- Remove noisy spectra with SNR deviating by more than 3σ from the mean $\rightarrow$ 95 spectra (before 101).
- Continuum normalization to flatten large-scale flux variations.
- Optional preprocessing steps:
 - Radial velocity correction using the relativistic Doppler formula:

$$\lambda_{\text{shifted}} = \lambda \cdot \sqrt{\frac{1 + v/c}{1 - v/c}} \quad \text{where} \quad v = v_{\text{RV}}$$

- Zero-mean normalization per spectrum: subtract mean flux.
- Unit-variance scaling per spectrum: divide by global SD.
- Region selection: specific spectral orders or wavelength intervals.
- All spectra are interpolated to a common wavelength grid.

Data Preprocessing: Overview

- Use telluric-corrected spectra.
- Remove noisy spectra with SNR deviating by more than 3σ from the mean $\rightarrow$ 95 spectra (before 101).
- Continuum normalization to flatten large-scale flux variations.
- Optional preprocessing steps:
 - Radial velocity correction using the relativistic Doppler formula:

$$\lambda_{\text{shifted}} = \lambda \cdot \sqrt{\frac{1 + v/c}{1 - v/c}} \quad \text{where} \quad v = v_{\text{RV}}$$

- Zero-mean normalization per spectrum: subtract mean flux.
 - Unit-variance scaling per spectrum: divide by global SD.
 - Region selection: specific spectral orders or wavelength intervals.
- All spectra are interpolated to a common wavelength grid.

Data Preprocessing: Overview

- Use telluric-corrected spectra.
- Remove noisy spectra with SNR deviating by more than 3σ from the mean $\rightarrow$ 95 spectra (before 101).
- Continuum normalization to flatten large-scale flux variations.
- Optional preprocessing steps:
 - Radial velocity correction using the relativistic Doppler formula:

$$\lambda_{\text{shifted}} = \lambda \cdot \sqrt{\frac{1 + v/c}{1 - v/c}} \quad \text{where} \quad v = v_{\text{RV}}$$

- Zero-mean normalization per spectrum: subtract mean flux.
- Unit-variance scaling per spectrum: divide by global SD.
- Region selection: specific spectral orders or wavelength intervals.
- All spectra are interpolated to a common wavelength grid.

Data Preprocessing: Overview

- Use telluric-corrected spectra.
- Remove noisy spectra with SNR deviating by more than 3σ from the mean $\rightarrow$ 95 spectra (before 101).
- Continuum normalization to flatten large-scale flux variations.
- Optional preprocessing steps:
 - Radial velocity correction using the relativistic Doppler formula:

$$\lambda_{\text{shifted}} = \lambda \cdot \sqrt{\frac{1 + v/c}{1 - v/c}} \quad \text{where} \quad v = v_{\text{RV}}$$

- Zero-mean normalization per spectrum: subtract mean flux.
 - Unit-variance scaling per spectrum: divide by global SD.
 - Region selection: specific spectral orders or wavelength intervals.
- All spectra are interpolated to a common wavelength grid.

Data Preprocessing: Overview

- Use telluric-corrected spectra.
- Remove noisy spectra with SNR deviating by more than 3σ from the mean $\rightarrow$ 95 spectra (before 101).
- Continuum normalization to flatten large-scale flux variations.
- Optional preprocessing steps:
 - Radial velocity correction using the relativistic Doppler formula:

$$\lambda_{\text{shifted}} = \lambda \cdot \sqrt{\frac{1 + v/c}{1 - v/c}} \quad \text{where} \quad v = v_{\text{RV}}$$

- Zero-mean normalization per spectrum: subtract mean flux.
 - Unit-variance scaling per spectrum: divide by global SD.
 - Region selection: specific spectral orders or wavelength intervals.
- All spectra are interpolated to a common wavelength grid.

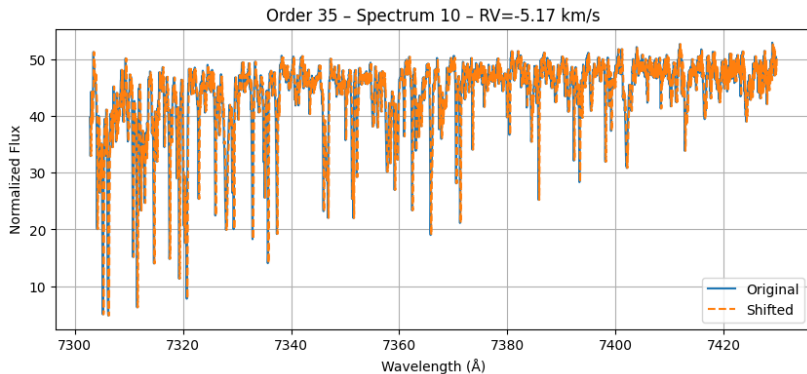
Data Preprocessing: Overview

- Use telluric-corrected spectra.
- Remove noisy spectra with SNR deviating by more than 3σ from the mean $\rightarrow$ 95 spectra (before 101).
- Continuum normalization to flatten large-scale flux variations.
- Optional preprocessing steps:
 - Radial velocity correction using the relativistic Doppler formula:

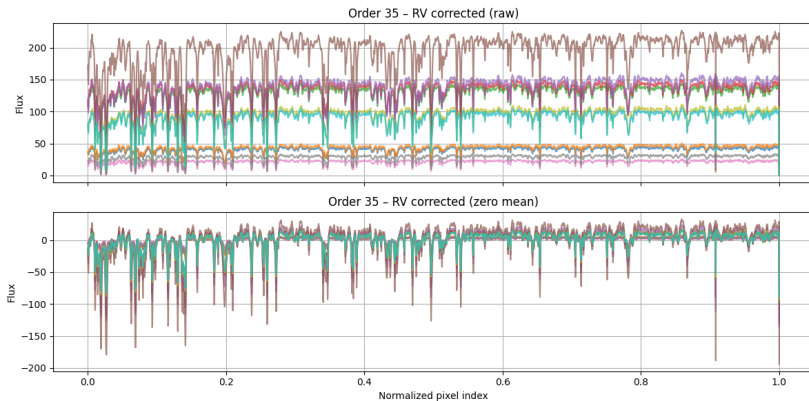
$$\lambda_{\text{shifted}} = \lambda \cdot \sqrt{\frac{1 + v/c}{1 - v/c}} \quad \text{where} \quad v = v_{\text{RV}}$$

- Zero-mean normalization per spectrum: subtract mean flux.
 - Unit-variance scaling per spectrum: divide by global SD.
 - Region selection: specific spectral orders or wavelength intervals.
- All spectra are interpolated to a common wavelength grid.

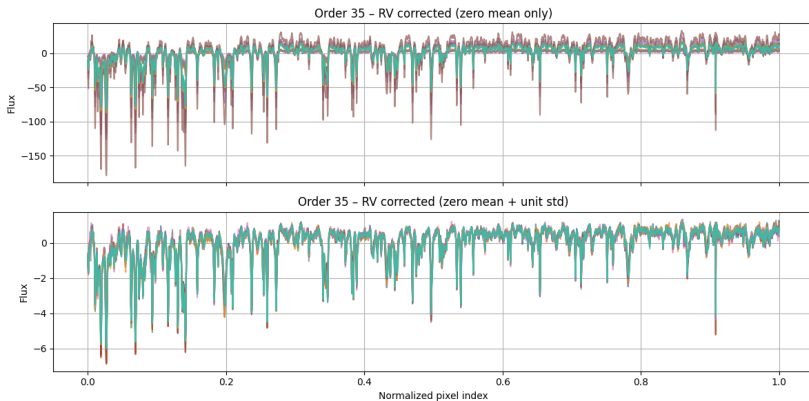
Data Preprocessing: RV Correction



Data Preprocessing: Zero Mean Normalization



Data Preprocessing: Unit Variance Scaling



Autoencoder Modeling

From Simple to Advanced Models

- **Step 1:** Simple Convolutional Autoencoder to capture key spectral structure.
- Step 2: Deepened architecture to learn fine-grained features.
- Step 3: Addressed SNR bias with decorrelation penalties applied only to some latent dimensions.
- Step 4: Introduced dropout for robustness and denoising variants.
- Step 5: Moved to Variational Autoencoder (VAE) framework for better latent space modeling.

From Simple to Advanced Models

- **Step 1:** Simple Convolutional Autoencoder to capture key spectral structure.
- **Step 2:** Deepened architecture to learn fine-grained features.
- **Step 3:** Addressed SNR bias with decorrelation penalties applied only to some latent dimensions.
- **Step 4:** Introduced dropout for robustness and denoising variants.
- **Step 5:** Moved to Variational Autoencoder (VAE) framework for better latent space modeling.

From Simple to Advanced Models

- **Step 1:** Simple Convolutional Autoencoder to capture key spectral structure.
- **Step 2:** Deepened architecture to learn fine-grained features.
- **Step 3:** Addressed **SNR bias** with decorrelation penalties applied only to some latent dimensions.
- **Step 4:** Introduced dropout for robustness and denoising variants.
- **Step 5:** Moved to **Variational Autoencoder (VAE)** framework for better latent space modeling.

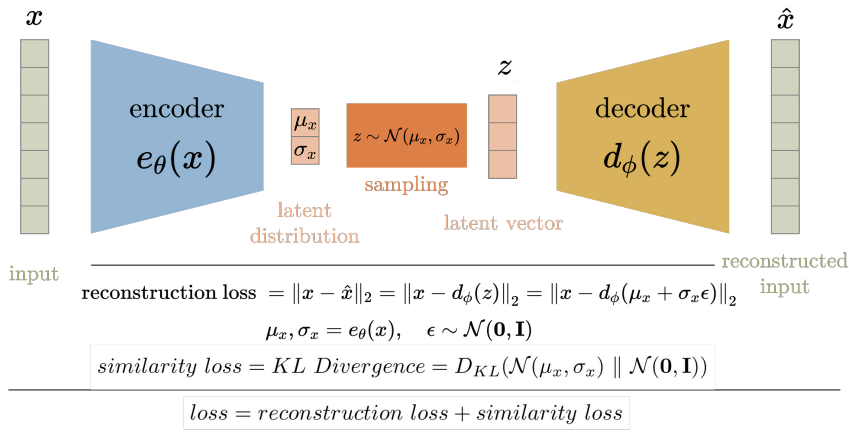
From Simple to Advanced Models

- **Step 1:** Simple Convolutional Autoencoder to capture key spectral structure.
- **Step 2:** Deepened architecture to learn fine-grained features.
- **Step 3:** Addressed **SNR bias** with decorrelation penalties applied only to some latent dimensions.
- **Step 4:** Introduced dropout for robustness and denoising variants.
- **Step 5:** Moved to Variational Autoencoder (VAE) framework for better latent space modeling.

From Simple to Advanced Models

- **Step 1:** Simple Convolutional Autoencoder to capture key spectral structure.
- **Step 2:** Deepened architecture to learn fine-grained features.
- **Step 3:** Addressed **SNR bias** with decorrelation penalties applied only to some latent dimensions.
- **Step 4:** Introduced dropout for robustness and denoising variants.
- **Step 5:** Moved to **Variational Autoencoder (VAE)** framework for better latent space modeling.

Variational Autoencoder



Implemented Architectures

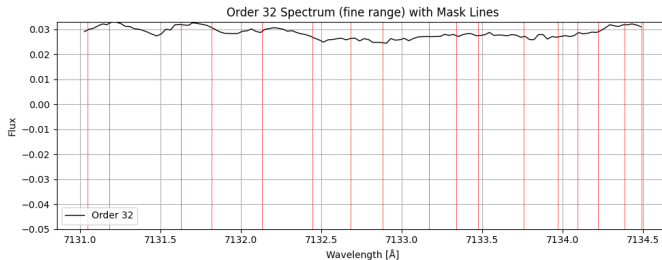
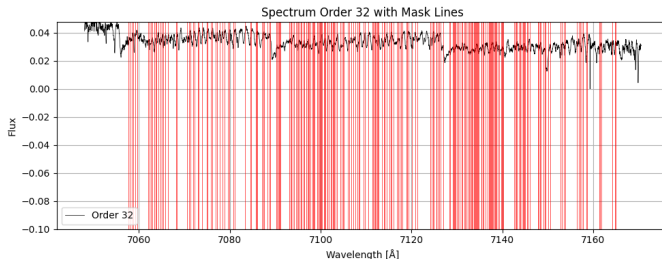
- **SimpleConvVAE1D:** basic convolutional encoder-decoder VAE with optional dropout.
 - **FullConvVAE1D:** deeper VAE with configurable residual blocks and dense layers.
 - **UNetVAE1D:** U-Net-style VAE using skip connections for more precise reconstructions.
-
- **Loss:** annealed over time.
 - **Dropout:** Monte Carlo dropout to increase robustness.

Noise Injection Strategy

- Add Gaussian noise with configurable standard deviation.
 - Generate X augmented spectra per original.
 - Optionally **protect mask lines** (absorption lines) from heavy perturbation.
-
- **Noise Protection:** based on proximity to masked absorption lines using Gaussian decay.

Example Configuration

Selected Input Region: Order 32 (7137.0 - 7140.5 Å)



→ Full Preprocessing (RV-shift, zero mean, unit variance)

Model Configuration and Training

- **Noise Injection Parameters:**

- NOISE_STD = 0.2, NOISE_FACTOR = 15
- SIGMA_PROTECT = 0.07, MIN_WEIGHT = 0.02

Model Settings

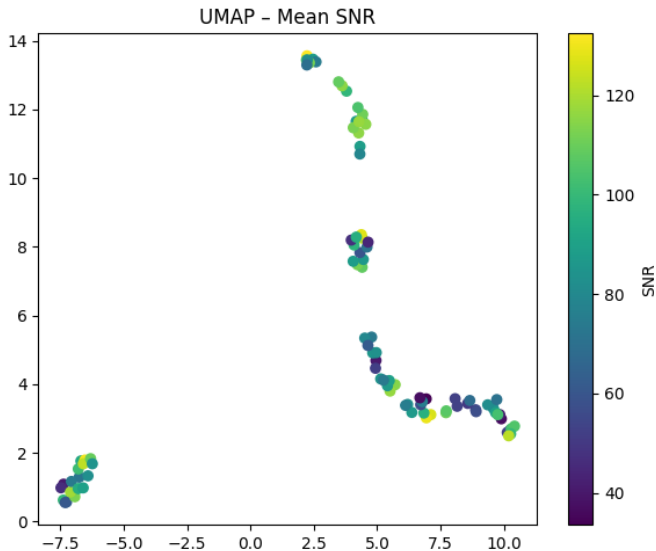
- **VAE Model:** FullConvVAE1D
- **Latent Dimension:** 16
- **Architecture:** Conv filters (32, 64), kernel size 3, fully symmetric
- **Extras:** Residual blocks, dense layer in the encoder and decoder, no MC dropout

- **Training Setup:**

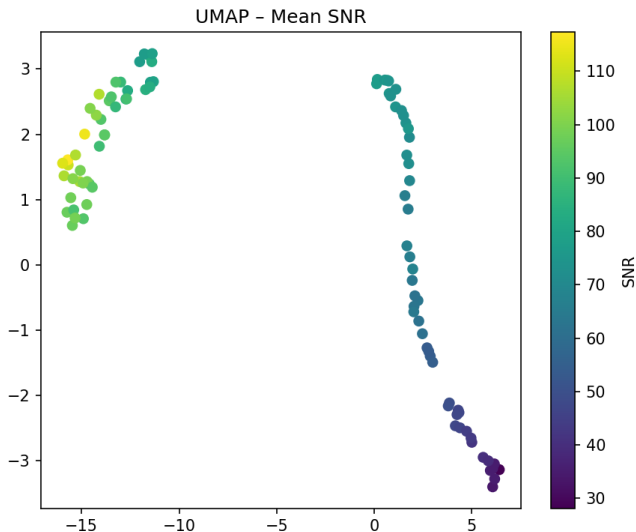
- EPOCHS = 5000, BATCH_SIZE = 95 (i.e., full batch),
LEARNING_RATE = $1e-5$
- KL annealing with BETA_MAX = $1e-5$; and warm-up

- **Denoising:** USE_DENOISING = True - target is clean input

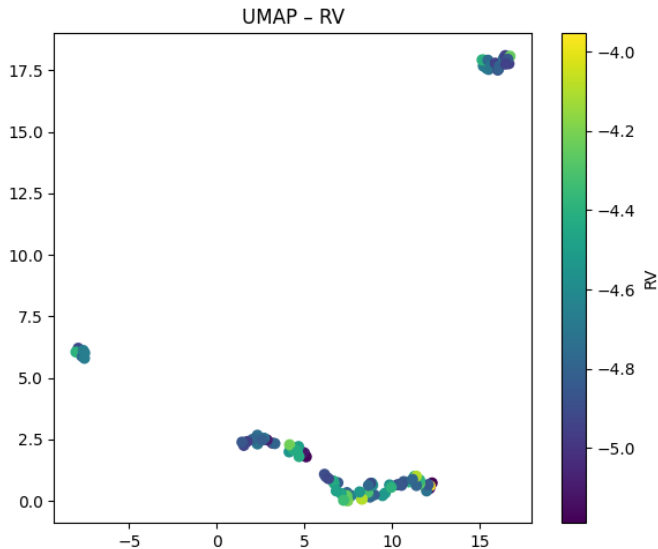
Signal to Noise Ratio (SNR)



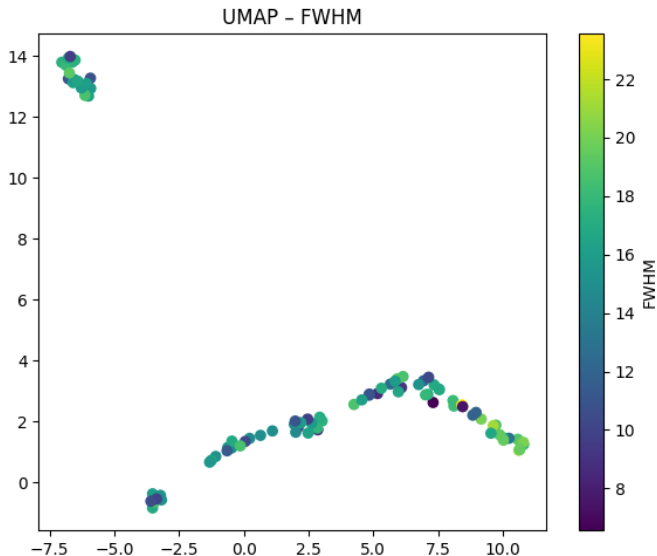
SNR-Comparison without RV Shift



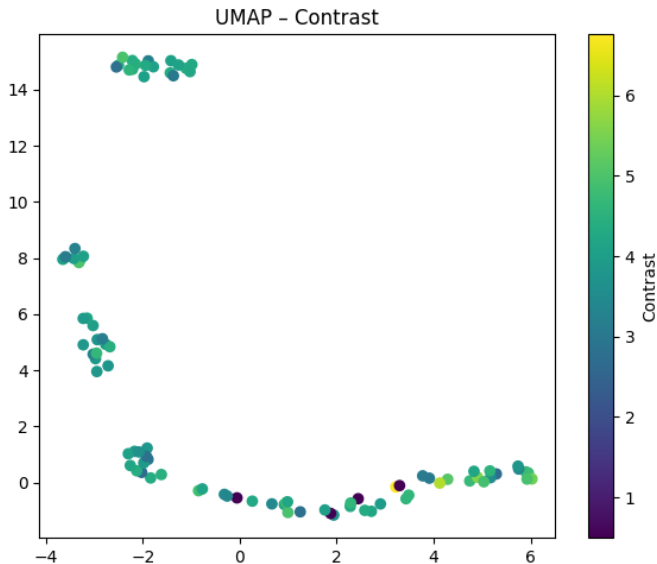
Radial Velocity (RV)



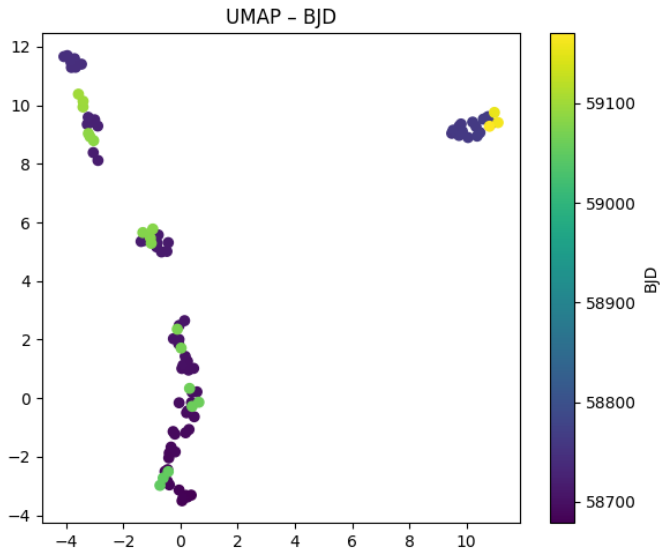
Full Width at Half Maximum (FWHM)



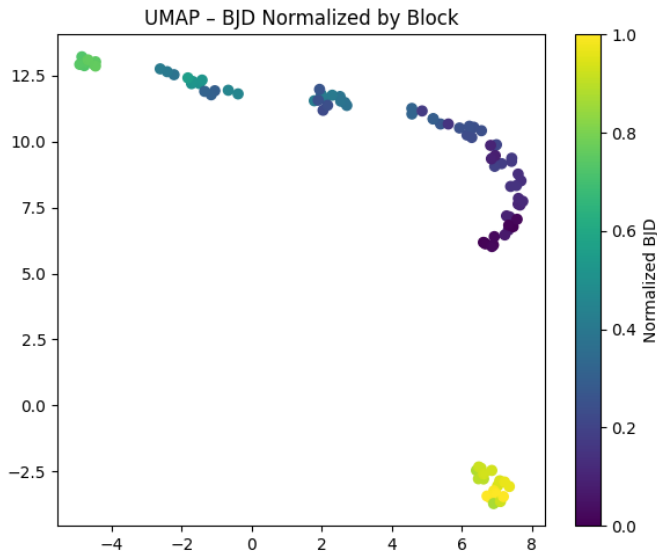
Contrast



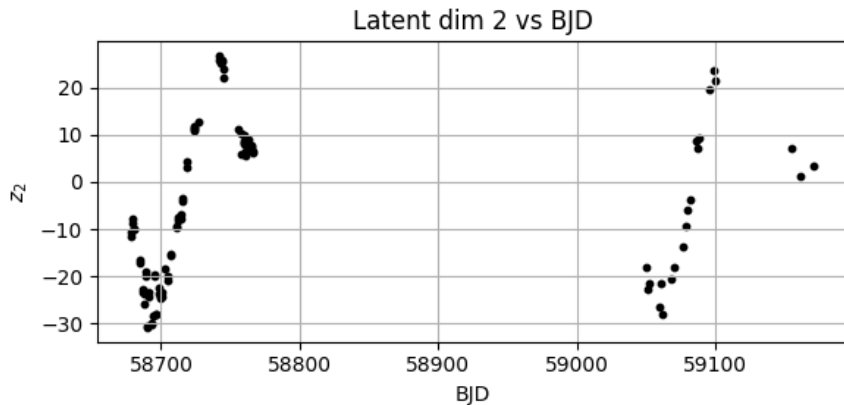
Barycentric Julian Date (BJD)



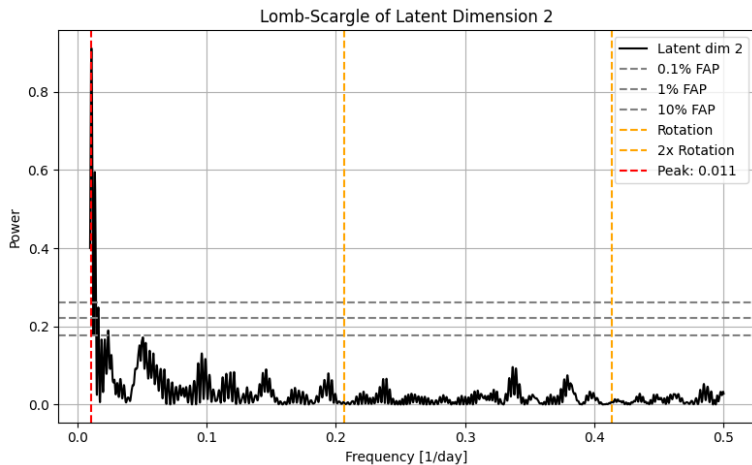
BJD with Color Normalization by Observation Block



Latent Dimension vs. BJD

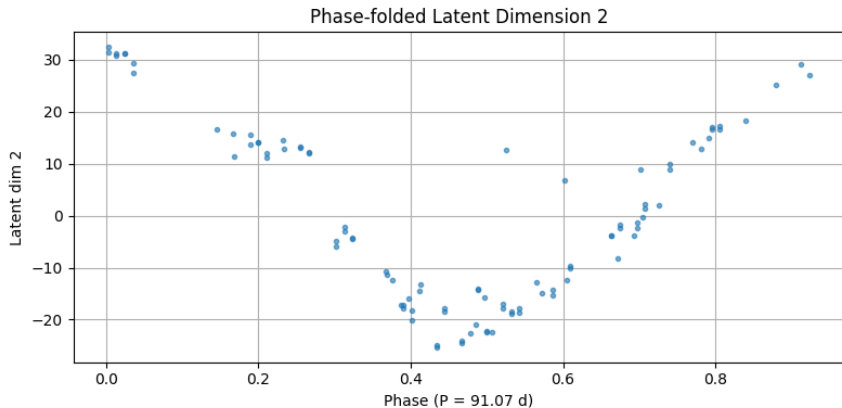


Lomb-Scargle Periodogram

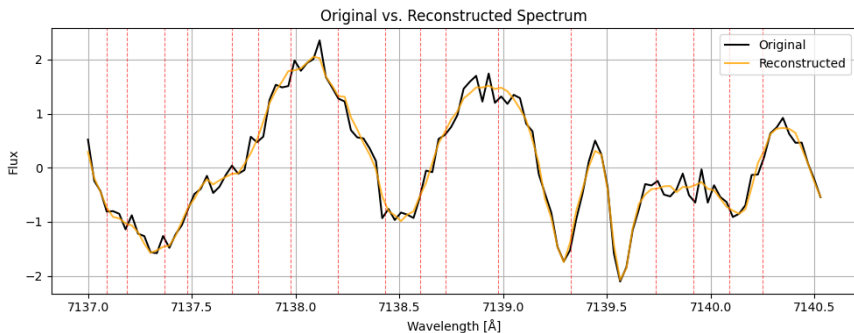


Period: **91.073 days**

Phase-Folded Latent Dimension



Reconstruction Performance



- Achieved significant reduction in SNR correlation.
- Unexplained time-dependent structure in latent space (≈ 90 days).
- *Ongoing work*: more data needed - potentially including solar spectra.

Thank you!

Questions?

Reconstruction Performance of UNetVAE1D

